

Omnivise iSER[®] Actuator

Intelligent Self-Contained Electrohydraulic Redundant Actuator

Increase reliability with a low-cost installation

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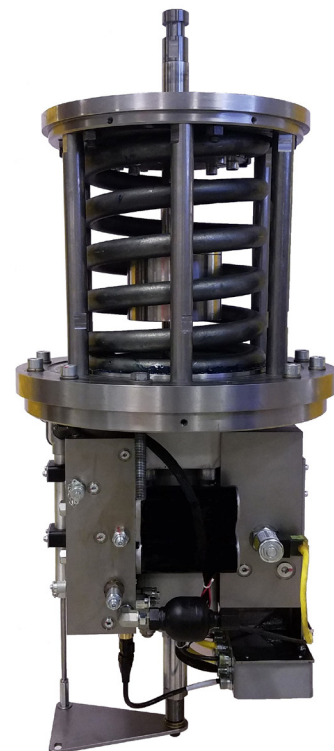
Siemens Energy Omnivise iSER Actuators are designed to precisely control the position of steam admission valves for power generation turbines. iSER actuators can be applied to any load, specifically for positioning, holding and trip applications. They meet the most stringent requirements for safety, reliability, and ruggedness.

Market-based benefits

- Increased reliability
 - If one motor or pump fails, it still operates thanks to built-in redundancies
- Lowered maintenance costs
 - Entire hydraulic system is self-contained, hence avoiding major oil leaks
 - PLC controls and onboard sensors allow for predictive instead of reactive maintenance
- Significantly reduced installation cost and schedule
 - Upgrade hydraulic controls without the need for high-pressure hydraulic piping installation
 - Up to 40% overall cost reduction

Specific Advantages

- Extremely accurate positioning and almost infinite power
- Eliminates the need for a dedicated and expensive hydraulic power unit
- Redundant pumps, motors, LVDTs, and trip solenoids remove single-point vulnerabilities
- Hydraulic system is self-contained with no external pipes or connections
- Easy connectivity to plant DCS via Profinet or Modbus TCP protocols



Actual product may differ

How it works: precision based modulating actuator

These actuator assemblies are double-acting. Motors and pumps provide hydraulic pressure to either side of the piston and check valves maintain the hydraulic pressure required to hold the piston in place when a desired position has been reached.

Linear Variable Differential Transformers (LVDT) provide position feedback for extremely accurate closed-loop control and near infinite positioning.

The actuator also has a modular trip system and return spring as a means for fail-safe turbine protection. In the event of an emergency, the trip system and return spring rapidly close the valve.

Key components

- ✓ High-pressure actuator assembly
- ✓ Redundant motor/pump assemblies (may be ordered with simplex or triplex motor/pumps)
- ✓ Control and trip manifolds with integral components (options available)
- ✓ Remote control panel with emergency stop button and HMI display
- ✓ Duplex LVDTs for feedback (may be ordered with simplex or triplex)
- ✓ Return spring for fail-safe shut down

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For more information,
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